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van't Hoff found out that for dilute solutions, the osmotic pressure is directly proportional to the molar concentration of the solute and the temperature of the solution. He proposed the following equation to calculate osmotic pressure which is now called as van't Hoff equation.

$$\pi = cRT \quad \text{.....9.31}$$

Here,

c = Concentration of the solution in molarity

T = Temperature

R = Gas constant

Determination of molar mass from osmotic pressure

According to van't Hoff equation

$$\pi = cRT$$

$$c = \frac{n}{V}$$

Here, n= number of moles of solute dissolved in 'V' litre of the solution.

Therefore, $\pi = \frac{n}{V} RT$ or

$$\pi V = nRT \quad (9.32)$$

If the solution is prepared by dissolving w_B g of nonvolatile solute in w_A g of solvent, then the number of moles 'n' is,

$$n = w_B / M_B$$

Since, M_B = molar mass of the solute

Substituting the 'n' in 9.32, we get,

$$\pi = \frac{w_B}{V} \frac{RT}{M_B}$$

$$M_B = \frac{w_B}{V} \frac{RT}{\pi} \quad (9.33)$$

From the equation 9.33, molar mass of the solute can be calculated.

Significances of osmotic pressure over other colligative properties

Unlike elevation of boiling point (for 1 molal solution the elevation in boiling point is 0.512 °C for water) and the depression in freezing point (for 1 molal solution the depression in freezing point is 1.86 °C for water), the magnitude of osmotic pressure is large.

The osmotic pressure can be measured at room temperature enables to determine the molecular mass of biomolecules which are unstable at higher temperatures.

Even for a very dilute solution, the osmotic pressure is large.

Isotonic solutions

Two solutions having same osmotic pressure at a given temperature are called isotonic solutions. When such solutions are separated by a semipermeable membrane, solvent flow between one to the other on either direction is same, i.e. the net solvent flow between the two isotonic solutions is zero.

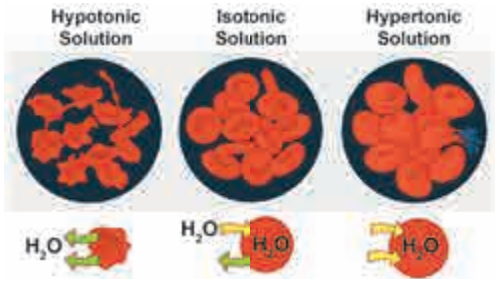


Figure 9.14 Isotonic solutions

The osmotic pressure of the blood cells is approximately equal to 7 atm at 37°C. The intravenous injections should have same osmotic pressure as that of the blood (isotonic with blood). If the Intravenous solutions are too dilute that is